SPARK ACADEMY

THE SPRINT AI TRAINING FOR AFRICAN MEDICAL IMAGING KNOWLEDGE TRANSLATION

2026 PROJECT CARD 3

Title: Improving glioma segmentation generalizability with African MRI representation

Research question
Does training or adapting glioma MRI models with BraTS-Africa data reduce performance bias when models are evaluated on Sub-Saharan African patients?

Why this is a real gap
BraTS models have historically been trained on datasets from North America and Western Europe. The BraTS-Africa dataset was created because models trained without African representation may not adequately delineate tumor regions in African populations and lower-resource imaging environments. The Radiology: Artificial Intelligence dataset paper states explicitly that including MRI data from African populations improves machine-learning model performance for African brain tumor segmentation.

Biological/radiological mechanism
The issue extends beyond population representation to acquisition context. BraTS-Africa MRIs come from 1.5-T clinical scanners in Nigerian institutions and reflect lower-resource image- quality conditions, protocol heterogeneity, and later disease presentation. These factors can shift tumor appearance, contrast, and artifact profiles compared with high-resource datasets.

LMIC clinical relevance
A model that performs well on global benchmark data but poorly on African scans is not clinically equitable. This project directly tests whether African data inclusion improves deployability in Sub-Saharan settings.

Objectives
1. Train a baseline model using non-African BraTS data where permitted.
2. Evaluate its performance on BraTS-Africa.
3. Train or fine-tune the same architecture using BraTS-Africa data.
4. Compare performance before and after African-domain adaptation.
5. Measure improvement separately for enhancing tumor, tumor core, and whole tumor regions.
6. Evaluate robustness using site-aware or scanner-aware splits if metadata are available.
7. Assess whether African-domain adaptation improves calibration and uncertainty beyond Dice score.

Dataset available
BraTS-Africa contains MRI data from 146 adult patients retrospectively acquired from multiple Nigerian institutions, with 95 presumed adult diffuse glioma cases and 51 other central nervous system neoplasm cases in the full public release. The dataset was specifically curated to expand BraTS representation to African populations and resource-constrained imaging settings. The BraTS-Lighthouse 2025 platform also confirms that BraTS now includes multiple challenge tasks under a broader benchmarking framework, including subregion segmentation and related AI evaluation tasks.

Why this is not just segmentation
The core question is domain generalization and equity: whether AI trained in high-resource settings can be adapted to work reliably for African patients.